RNA FISH

Considerations:

- Use #1.5H high precision coverglasses from Marienfeld
- The volumes in this protocol are optimized for 18 mm round coverslips. Scale volumes up or down according to the size of your coverslips.
- This protocol is optimized for 40-mer FISH probe (20-30 -mer long probes require different washing conditions)
- **Ensure that all reagents are RNase-free**
- **Clean tweezers using RNaseZap, then rinse with RNase-free water and 100% EtOH**

Day 1 - PFA fixation

1. Rinse coverslips quickly with 1× PBS with Ca^{2+} and Mg^{2+} pre-warmed to 37°C twice (*the addition of Ca^{2+} and Mg^{2} helps preserving cell adhesion and membrane integrity. If cell density is high regular PBS is also acceptable*)
2. Replace with 1× PBS containing 4% PFA at room temperature (RT) and incubate for 10 min
3. Discard the PFA solution and rinse with 1× PBS/125 mM Glycine at RT
4. Replace with fresh 1× PBS/125 mM Glycine and incubate at RT, 5 min
5. Wash with 1× PBS twice at RT
6. Rinse with ice-cold 70% ethanol (*dispense the solution onto the walls of the plate rather than directly onto the coverslips*)
7. Replace with fresh ice-cold 70% ethanol and store at 4 °C

   Note: Cells can be stored in 70% ethanol at 4 °C for several months. If proceeding immediately with probe hybridization, the 70% ethanol incubation can instead be performed at RT for 15 min.

Day 2 - RNA FISH

Note: Pre-warm FISH wash buffer to RT and avoid light

1. Wash with 2× SSC at RT twice
2. Wash twice with FISH wash buffer at RT
3. Equilibrate in FISH wash buffer for at least 5 min
4. Prepare the RNA hybridization mix:
   - 99 μl of the FISH RNA hybridization buffer
   - 1 μl of 100 nM RNA FISH probe to reach final concentration of 1 nM per oligo

     Note: Optimize probe concentration according to your probe set
   - Vortex briefly and spin down
5. Dispense the RNA hybridization mix onto the parafilm placed inside a humidity chamber and place the coverslip with cells facing down on the hybridization mix
6. Seal the humidity chamber with parafilm and incubate at 37 °C overnight

Day 3 – Fluorescent oligo hybridization

Note: Pre-warm FISH wash buffer to RT and avoid light. The buffer may be left at RT in the dark overnight.

1. Set water bath to 65 °C and pre-warmed 0.2× SSC/0.2% Tween-20
2. Transfer the coverslip to a plate filled with 2× SSC/0.2% Tween-20
3. Rinse with 2× SSC/0.2% Tween-20
4. Wash twice with pre-warmed 0.2× SSC/0.2% Tween-20 at 65 °C by incubating the plate on the surface of water bath 10 min (*see note 2 below*)
5. Wash twice with 2× SSC at RT
6. Wash twice with FISH wash buffer at RT
7. Equilibrate in FISH wash buffer for at least 5 min
8. Prepare the FISH Readout hybridization mix:
    - 99 μl of the FISH Readout hybridization buffer
    - 1 μl of 2 μM fluorescently-labelled oligo to reach final concentration of 20 nM
      Note: Avoid exposure to light!
    - Vortex briefly and spin down
9. Dispense the readout hybridization mix onto the parafilm placed inside a humidified chamber and place the coverslip with cells facing down on the hybridization mix
10. Seal the humidified chamber with parafilm and incubate at 30 ºC overnight, avoid light *(this incubation can be reduced to as little as 2 h and proceed immediately with day4)*

Note 2: If probes are 20-30 mers long or if the signal is weak, perform post-hybridization washes in FISH WASH buffer at 37°C for 30 min twice. Coverslips can be transferred directly into a 12-well plate containing FISH WASH buffer. Washes may be performed in the same oven used for probes hybridization. After washing, proceed directly to Step 5.

Day 4 – Mounting and imaging

1. Set water bath to 45 °C and pre-warmed 0.2× SSC/0.2% Tween-20
2. Transfer the coverslip to a plate containing 2× SSC/0.2% Tween-20
3. Rinse with 2× SSC/0.2% Tween-20
4. Wash twice with pre-warmed 0.2× SSC/0.2% Tween-20 by incubating the plate on the surface of water bath at 45 °C, 10 min *(see note below)*
5. Wash twice with 2× SSC at RT
6. Incubate with 0.1 ng/μl Hoechst 33342/2× SSC at RT, 30 min
7. Wash twice with 2× SSC at RT
8. Mount with Prolong Gold antifade mounting medium (using 7μl per 18mm coverslip). Allow to dry the O/N in the dark and then seal with Fixogum for imaging. If GLOX mounting medium is used, imaging may be performed immediately.

Note: If probes are 20–30 mers long or if the signal is weak, perform post-fluorescent oligo hybridization washes in 2X SSC/ 0,2% tween20 2x 8 min. Coverslips can be transferred directly into a 12-well plate containing 2X SSC/0,2% tween20. Washes may be performed in the same oven used for secondary fluorescent oligos hybridization. After washes, proceed to Step 5.

Reagents:

20 mg/ml tRNA

Dissolve 99.5 mg (2,000 units) of tRNA in 4,970 µl of nuclease-free water to obtain a solution at 20 mg/ml.
Aliquot in 1 ml and store at –20 °C up to year or –80 °C indefinitely.

40% (wt/vol) Glucose

Dissolve 20 g of glucose in 50 ml of nuclease-free water. Filter the solution using a 0.2-µm syringe filter
and store it at 4 °C for up to a year.

50 mg/ml Glucose oxidase

Dissolve 5 mg enzyme powder in 1 ml of 50 mM sodium acetate pH 5.0. Aliquot the solution in 10 µl and store them at –20 °C for up to several months.

200 mM Trolox

Dissolve 50 mg of the Trolox powder in 1 ml of pure ethanol. Divide the solution into 50- to 100-µl portions and store them at –20 °C for up to several months.

10% (v/v) Triton X-100

Dilute 1 ml of 100% Triton X-100 to 9 ml nuclease-free water. Mix by rotation or inverting until all Triton X-100 dissolve. Store it at RT up to 3 months.

10% (v/v) Tween-20

Dilute 1 ml of 100% Tween-20 to 9 ml nuclease-free water. Mix by rotation or inverting until all Tween-20 dissolve. Store it at RT up to 3 months.

1.25 M Glycine

Dissolve 4.68 g glycine in 50 ml nuclease-free water. Mix by rotation or inverting until all powder dissolves. Store it at RT for up to 6 months.

FISH wash buffer
(25% Formamide, 2× SSC)

Reagents	Volume
100% Formamide (Ambion #AM9342)	12.5 ml
20× SSC (Ambion #AM9765)	5 ml
Nuclease-free water (Ambion #AM9932)	32 ml
Total	50 ml

Store at 4 °C up to a few months. **Bring it to RT before use.**

FISH RNA hybridization buffer
(2× SSC, 10% Dextran sulfate, 25% Formamide, 1 mg/ml E.coli tRNA, 0.2 mg/ml BSA, 2 mM RVC)

Reagents	Volume
20× SSC (Ambion #AM9765)	1 ml
50% Dextran sulfate solution (Sigma #S4030)	2 ml
100% Formamide (Ambion #AM9342)	2.5 ml
20 mg/ml *E.coli* tRNA (Sigma #R1753)	500 *µl*
50 mg/ml BSA (Ambion #AM2616)	40 *µl*
200 mM RVC (NEB #S1402S)	100 *µl*
Nuclease-free water (Ambion #AM9932)	3.86 ml
Total	10 ml

Preparation of the FISH RNA FISH hybridization buffer

1. Add nuclease-free water and 20× SSC to a 15 ml tube
2. Add dextran sulfate solution and mix well to dissolve it. If necessary, warm the solution at 37 °C *(alternatively add the indicated volume of the 50% solution and warm it to better dissolve)*
3. Add other components to the tube and mix well. Make sure there is no precipitate
   Note: Be sure to completely dissolve RVC before adding
4. Check the pH and make sure it is between 7-8
5. Aliquot in 1 ml and store at -20 °C

FISH Readout hybridization buffer
(2× SSC, 10% Dextran sulfate, 25% Formamide)

Reagents	Volume
20× SSC (Ambion #AM9765)	1 ml
50% Dextran sulfate solution (Sigma #S4030)	2 ml
100% Formamide (Ambion #AM9342)	2.5 ml
Nuclease-free water (Ambion #AM9932)	4.5 ml
Total	10 ml

Preparation of the FISH Readout hybridization buffer

1. Add nuclease-free water and 20× SSC to a 15 ml tube
2. Add dextran sulfate solution and mix well to dissolve it. If necessary, warm the solution at 37 °C *(alternatively add the indicated volume of the 50% solution and warm it to better dissolve)*
3. Add other components to the tube and mix well. Make sure there is no precipitate.
4. Check the pH and make sure it is between 7-8
5. Aliquot in 1 ml and store at -20 °C

Equilibrium buffer (EQ buffer)
(10 mM Tris-HCl pH 7.5, 2× SSC, 0.4% Glucose, pH 7-8)

Reagents	Volume
1M Tris-HCl (pH 7.5) (Invitrogen 15567027)	100 µl
20× SSC (Ambion #AM9765)	1 ml
40% Glucose (Sigma #49139)	100 µl
Nuclease-free water (Ambion #AM9932)	8.8 ml
Total	10 ml

Store at RT up to a year.

Mounting medium (GLOX buffer)
(10 mM Trolox, 500 µg/ml Glucose Oxidase, 100 µg/ml Catalase, in EQ buffer)
Note: Prepare fresh before imaging. Add Trolox first and mix well before adding the enzyme
Do not use the Glucose Oxidase after three freeze-thaw cycles. Prepare small aliquots

Reagents	Volume
Equilibrium buffer (EQ buffer)	93 µl
200 mM Trolox (Sigma #238813)	5 µl
50 mg/ml Glucose Oxidase (Sigma #G2133)	1 µl
10 mg/ml Catalase (Sigma #C3515)	1 µl
Total	100 µl